Advanced Problem-Solving Lab – High-IQ Track

Activity H1 – Solve the "Hidden Problem" Behind the Stated Problem

Title: From Surface Problem to Real Problem

Goal

Train yourself to see the *real* problem, not just the stated one.

Instructions

You are given a "client statement":

"Our exam system is too slow. We want a Python script to speed it up."

1. Identify at least 3 possible *real* problems this statement might hide.
   - Hidden problem 1: Inefficient algorithm.
   - Hidden problem 2: High latency due to network / database access
   - Hidden problem 3: Resource limitation
2. For each hidden problem, state:
   - What data would you need to confirm it?
     - Problem 1: Execution time profiling of functions, algorithm analysis.
     - Problem 2: Network request logs
     - Problem 3: System resource usage stats.
3. Choose ONE hidden problem and re-frame it as a clear technical problem statement (2–3 lines):

New problem statement:

The exam system performance is degraded because its core processing algorithm has quadratic time complexity.

4. Design a very small Python experiment that could validate or reject your reframed problem (pseudo-code is fine):

python

 # Experiment idea (what will you measure, on what data, how?)

```
def original_algorithm (data):
    result = 0
for i in range (len(data)):
    for j in range (len (data)): result += data[i]* data[j]
    return result.

def optimized_algorithm (data):
    return sum(data)** 2.
```

Activity H2 – Multiple Competing Solutions

Title: Design 3 Solutions Before Coding 1

Goal

Avoid fixation. Explore at least three distinct solution strategies.

Instructions

Problem:

"You receive a stream of log lines (text). You must detect 'suspicious' activity in real time."

1. Give three different solution ideas, each using a distinct main technique or data structure.
   - Approach A (e.g., rule-based / regex / thresholds):

     apply thresholds for severity.
   - Approach B (e.g., statistical / sliding window / counters):

     Maintain a sliding window of recent log events
   - Approach C (e.g., simple ML / anomaly detection concept):

     Train a lightweight anomaly-detection algorithm
2. For each approach, list pros and cons (at least one each):
   - Approach A – Pros: Fast Cons: require manual rule maintenance
   - Approach B – Pros: capture temporal pattern Cons: Needs tuning of window, size & threshold
   - Approach C – Pros: Learns complex patterns Cons: requires training data
3. Choose one approach to implement first and justify:

I choose approach A *because* it provides quick, transparent detection with minimal setup & is suitable for real time processing of text logs

4. Sketch high-level Python structure (modules/classes, not full code):

python

 # Modules / classes / key functions list

```
def parse_line(self, line):
    pass
def match_rules(self, fields):
    pass
class rule engine:
    def load_rules(self):
        pass
    def evaluate(self, log_data):
        return is_suspicious
```

```
def real_time_monitor():
    parser = log parser()
    engine = Rule engine()
    alert = Alert system()
for log_line in log_stream:
    data = parser.parse_line(log_line)
    if engine.evaluate(data):
        alert.trigger_alert(log_line)
```

Activity H3 – Self-Constrained Challenge: "Elegant Code Only"

Title: Solve with Elegance (Clarity, Minimalism, Correctness)

Goal

Produce code that is not only correct, but also minimal and beautiful.

Problem

Implement a mini "pipeline" for processing a list of tasks:

- Each task has an ID, description, and status ("todo", "doing", "done").
- You must:
  - Load tasks from a JSON file.
  - Filter by status.
  - Sort by ID.
  - Print a clean summary.

Constraints (you impose on yourself)

1. Maximum 25 lines of *real* Python (excluding blanks and comments).
2. No function longer than 7 lines.
3. Names must be meaningful and short.

Steps

1. Design your data model in 2–3 lines:

_____

2. Decide at least one thing you will *not* do to keep it minimal:

_____

3. Write your code here (respecting constraints):

python

 # Elegant pipeline solution

4. Reflection:
   - What did you have to sacrifice (if anything) to keep elegance?

~~to keep code minimal & elegant & modular input validation~~
~~focusing on core operation within the strict line limit~~

```
import json.
Task = lambda d : {'id': d['id'], 'desc': d['desc'], 'status': d['status']}
for t in ts : print(f"{t['id']} : {t['desc']} ({t['status']})")
tasks = load ('tasks.json')
filtered = filter_by_status (tasks, 'todo')
sorted_tasks = sort_by_id (filtered)
summary (sorted_tasks)
```

Activity H4 – Complexity and Scaling Thought Experiment

Title: How Will Your Solution Break at Scale?

Goal

Develop realistic intuition about performance and scaling.

Instructions

Take ANY non-trivial function or algorithm you have recently written (search, summarization, scoring, etc.).

1. Describe what it does, in *one line*:

Merge sort is a divide & conquer algorithm.

2. Estimate its time complexity in Big-O notation, and justify in 1–2 lines:
   - Big-O: O(nlogn)
   - Because: Merge Sort divides the array into halves
3. Scaling scenarios – imagine input sizes:
   - $n = 10^3$ (1,000)
   - $n = 10^6$ (1,000,000)
   - $n = 10^9$ (1,000,000,000)

For each, predict: will it still be usable (Y/N) and why?

n	Usable (Y/N)	Reason (1 short sentence)
10^3	Y	Sorting 1000 elements is relatively fast
10^6	Y	still manageable with modern computing power
10^9	N	~~usable~~ sorting a billion elements with O(n log n) complexity

4. Suggest one change to improve scalability and briefly describe how it helps:

use a parallel sorting algorithm or distribute the sorting task across multiple processors to reduce overall time complexity.

Activity H5 – Adversarial Thinking: Break Your Own Design

Title: Attack Your Own Program

Goal

Develop robustness and defensive mindset.

Instructions

Choose one of your own projects or mini-tools (file reader, API client, CLI, etc.).

1. Write a short description of what it is supposed to do:

program reads a file specified by the user & displays the contents.

2. Now wear the hat of an *attacker* or malicious user.
   List 5 ways to misuse / break it:
   - Attack 1: providing a non-existent file path
   - Attack 2: feeding a very large file to cause memory issues
   - Attack 3: Using a file with special characters.
   - Attack 4: Attempting to read a dictionary.
   - Attack 5: Trying to exploit path.
3. For each attack, suggest one defence in plain language:
   - Defence for attack 1: check if the file exists
   - Defence for attack 2: Implement chunked reading.
   - Defence for attack 3: Handle encoding & special characters.
   - Defence for attack 4: Check if path is file, not a directory
   - Defence for attack 5: validate & sanitize input path.
4. Implement at least one of these defences in your code and paste the updated snippet:

python

 # Hardened code snippet:

```
import os
def safe_read_file(file_path):
    if not os.isfile(file_path):
        print("Error: Not a valid file")
    return.
    try:
        with open(file_path, 'r', encoding='utf-8') as file:
        for line in file:
            print(line.strip())
    except exception as e:
        print(f"An error occurred: {e}")
```

Activity H6 – Cross-Domain Transfer

Title: Apply Known Pattern to a New Domain

Goal

Practice lifting a known idea/pattern from one domain to a different one.

Instructions

Pick a pattern you already know well, e.g.:

- Sliding window
- Two-pointer technique
- Dynamic programming table
- Producer–consumer
- Map–reduce style
- Pipeline of filters

1. Name of pattern: Two - pointer technique
2. Original domain where you learned it (e.g., arrays, strings):

Arrays.

3. New domain to apply it in (e.g., logs, time series, learning activities, medical data):

Medical data

4. Describe how you would adapt the pattern to this new domain, in 5–8 lines (can include pseudo-code):

Use two pointers to traverse a sorted list of appointment start pointer 1 tracks the start of an appointment, pointer 2 tracks the end when pointer 1's start time < pointer 2's end time

5. Optional: implement a small prototype showing this transfer:

python

```
# Pattern transfer prototype

appointments = [(start1, end1), (start2, end2), ...]
Sort appointments by start time.
pointer 1 = 0
pointer 2 = 1
while pointer2 < len(appointments):
    if appointment[pointer 1][1] > appointments[pointer2][0]:
        print("overlap found")
    pointer 1+=1
    pointer 2+=1
```

Activity H7 – Meta-Cognition: Your Personal Problem-Solving Manual

Title: Write Your Own "How I Solve Problems" Guide

Goal

Turn your tacit skills into an explicit, reusable system.

Instructions

1. List the first three things you usually do when you get a new problem (be honest):
   - Step 1: panic & Google it.
   - Step 2: Try random approches
   - Step 3: Ask a friend.
2. Now, design your ideal 6–8 step process (what a "future you" would always do):

1. Define problem clearly
2. Break it into smaller parts
3. Research known solutions
4. Sketch potential approaches
5. prototype & test
6. Refine based on feedback
7. (optional) Document the solution
8. (optional) Share with others

3. For each step, write one question you will ask yourself:
   - Step 1 question: what exactly is the problem asking?
   - Step 2 question: Can I isolate the core issue?
   - Step 3 question: Has anyone solved something similar?
   - Step 4 question: what are the simplest possible solutions?
   - Step 5 question: Does my prototype work as expected?
   - Step 6 question: what can I improve?
4. Final instruction to yourself (one sentence):

"When I feel stuck, I will take a short." break & revisit the problem definition"